University of Florida, Department of Mathematics

Topology PhD Exam, January 2001

Work the following problems and show all work. Support all statements to the best of your ability.

1. Show that there is a continuous function from I onto I^2.

2. State and prove the Brouwer Fixed Point Theorem. You may assume that $H_n(S^n;\mathbb{Z}) = \mathbb{Z}$ for all $n \geq 1$.

3. Show that $H_n(S^n;\mathbb{Z}) = \mathbb{Z}$ and that $H_0(S^n;\mathbb{Z}) = \mathbb{Z}$ for $n \geq 1$.

4. Show that the fundamental group of S^n is trivial for $n > 1$.

5. Show that $\pi_1(P_n) = \mathbb{Z}_2$ for $n > 1$, where P_n is n-dimensional real projective space.

Answer the following with complete definitions or statements or short proofs.

6. State the Seifert–van Kampen Theorem.

7. State the Hahn–Mazurkiewicz Theorem.

8. State the Lefschetz Fixed Point Theorem for compact simplicial complexes.

9. Define the cone of a space X. Define the suspension of X.

10. State the exact homology sequence for a pair of spaces.

11. State the Jordan Curve Theorem.

12. State the Contraction Mapping Theorem.

13. State the Ascoli (Arzelà–Ascoli) Theorem.

14. Let A and B be disjoint closed sets in a metric space X. Show that there is a continuous function $f : X \to [0,1]$ such that $f^{-1}(1) = A$ and $f^{-1}(0) = B$.

15. Show that I and I^2 are not homeomorphic. Show that I^2 and I^3 are not homeomorphic.